Sales Analytics Report (Online Retail)

Generated on 2026-02-28

Executive Summary

• Dataset: 541,909 rows raw, 524,878 rows after cleaning (duplicates removed, cancellations/returns and invalid prices excluded).

• Time period: 2010-12-01 to 2011-12-09.

• Total revenue: 10,642,110.80 across 19,960 orders. Average order value: 533.17.

• United Kingdom contributes 84.6% of revenue.

• Customer value is concentrated: 'Champions' segment represents 1,317 customers and 73.0% of revenue.

• Forecasting (ETS weekly seasonality) suggests the next 30 days average daily revenue around 106,393 with typical uncertainty of ±29,637.

Core KPIs

KPI	Value
Total Revenue	10,642,110.80
Total Orders (Invoices)	19,960
Total Items Sold (Qty)	5,572,420
Unique Customers (with ID)	4,338
Countries	38
Average Order Value	533.17

Advanced Data Cleaning

Key cleaning rules applied:

• Removed exact duplicate rows (5,268).

• Flagged and excluded cancellations/returns (InvoiceNo starting with 'C' or Quantity <= 0).

• Excluded invalid price rows (UnitPrice <= 0).

• Parsed InvoiceDate to datetime and derived calendar features (Year, Month, Day of Week, Hour).

Notes:

• CustomerID is missing in 24.9% of raw rows; customer-level analyses use rows with CustomerID present.

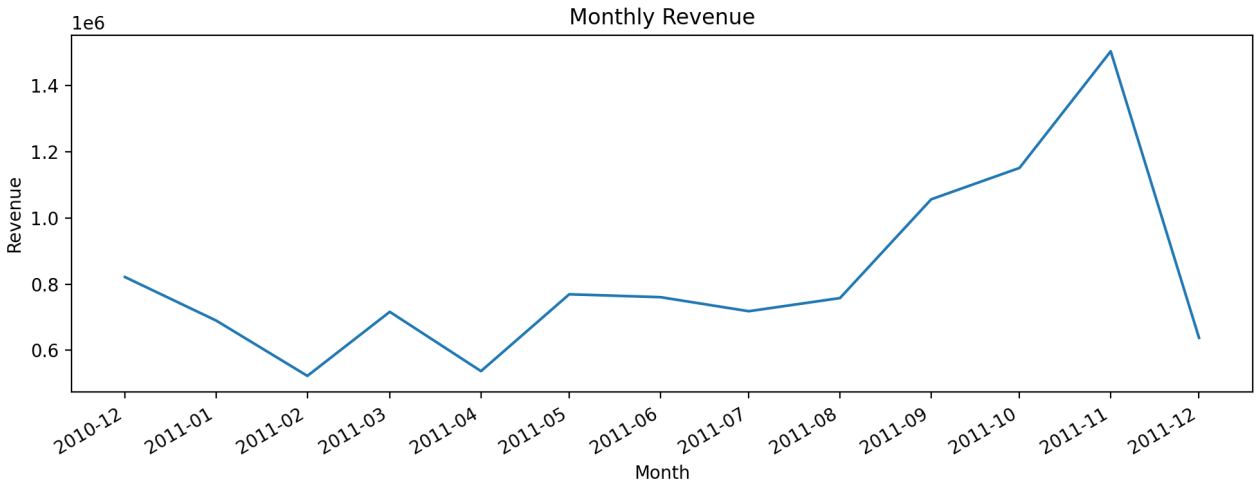
Descriptive Statistics

Metric	Quantity	UnitPrice	Revenue
count	524878.0	524878.0	524878.0
mean	10.62	3.92	20.28

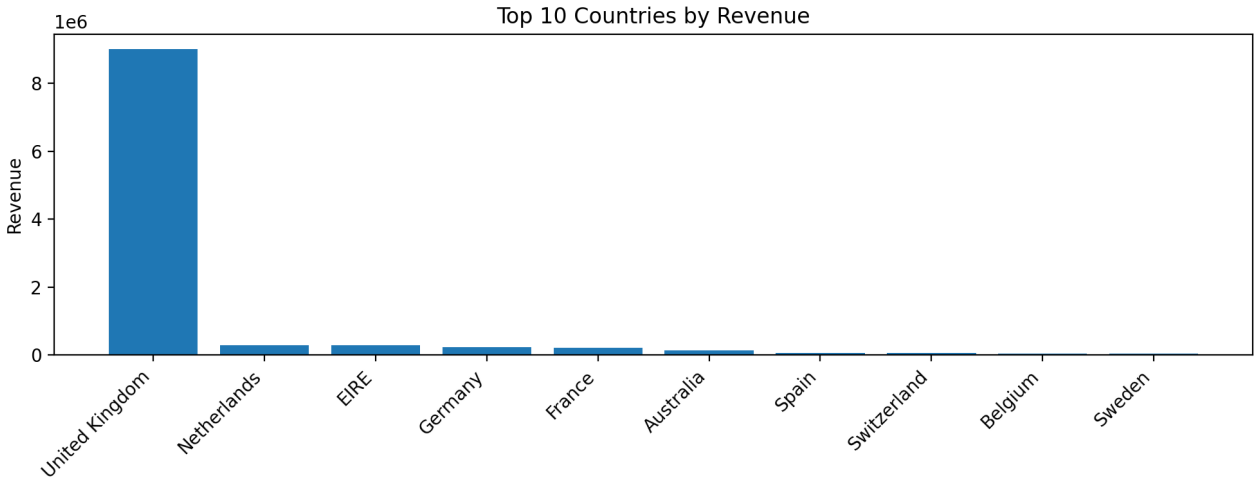
Metric	Quantity	UnitPrice	Revenue
std	156.28	36.09	271.69
min	1.0	0.0	0.0
25%	1.0	1.25	3.9
50%	4.0	2.08	9.92
75%	11.0	4.13	17.7
max	80995.0	13541.33	168469.6

Visual Trends & KPIs

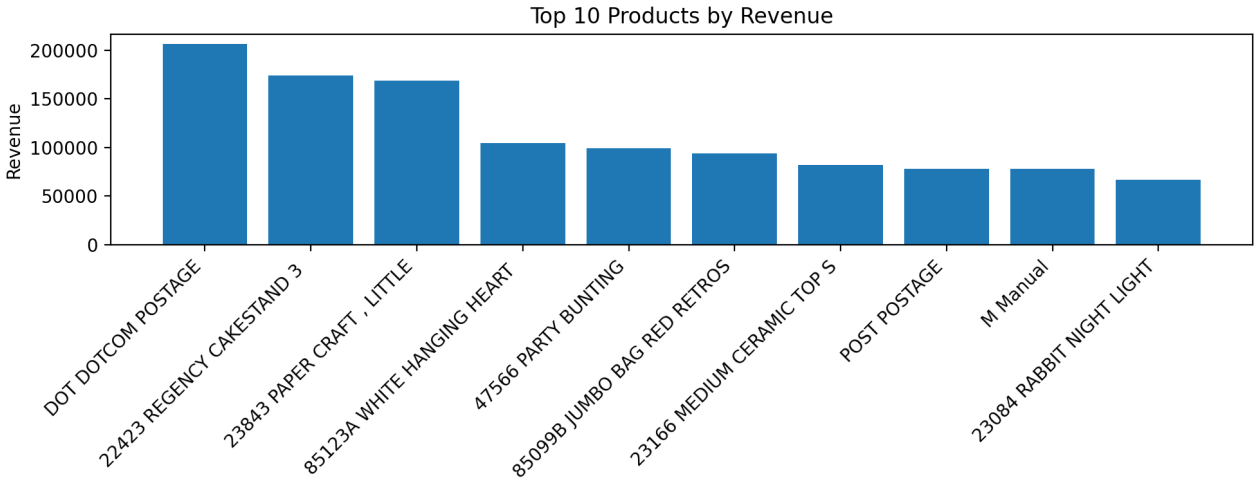
Monthly revenue trend



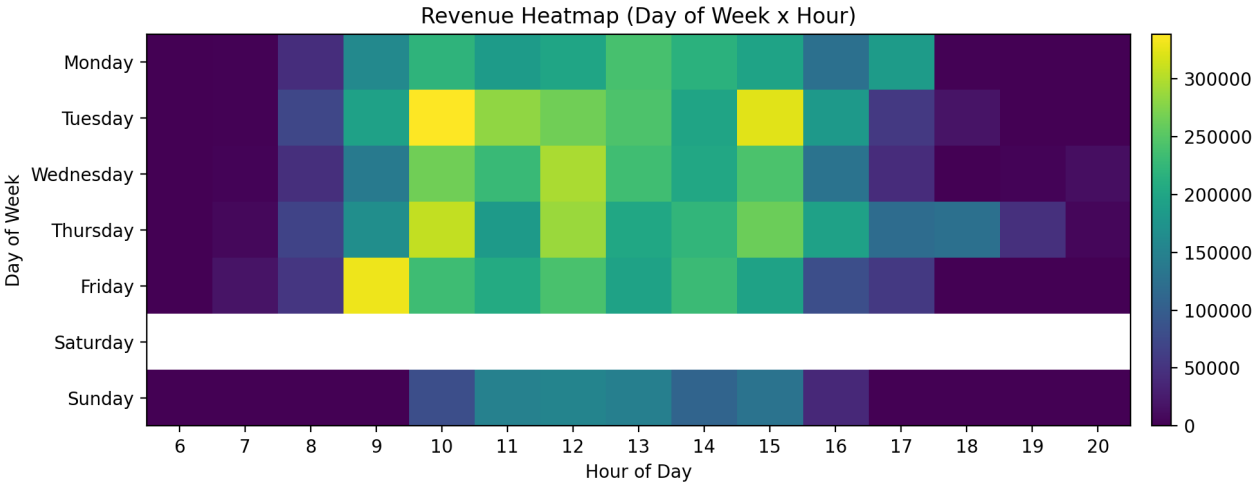
Revenue concentration by country (Top 10)



Top products by revenue (Top 10)



When revenue happens: day-of-week x hour heatmap



Customer Value Segmentation (RFM)

RFM is computed for customers with CustomerID using the last invoice date as the reference point. Segments are heuristic groupings of Recency, Frequency and Monetary value.

Segment	Customers	Revenue	Revenue Share
Champions	1,317	6,488,695	73.0%
Need Attention	759	952,844	10.7%
Loyal	442	470,040	5.3%
At Risk	565	455,081	5.1%
Hibernating	863	403,400	4.5%
New/Promising	392	117,149	1.3%

Predictive Analytics / Forecasting

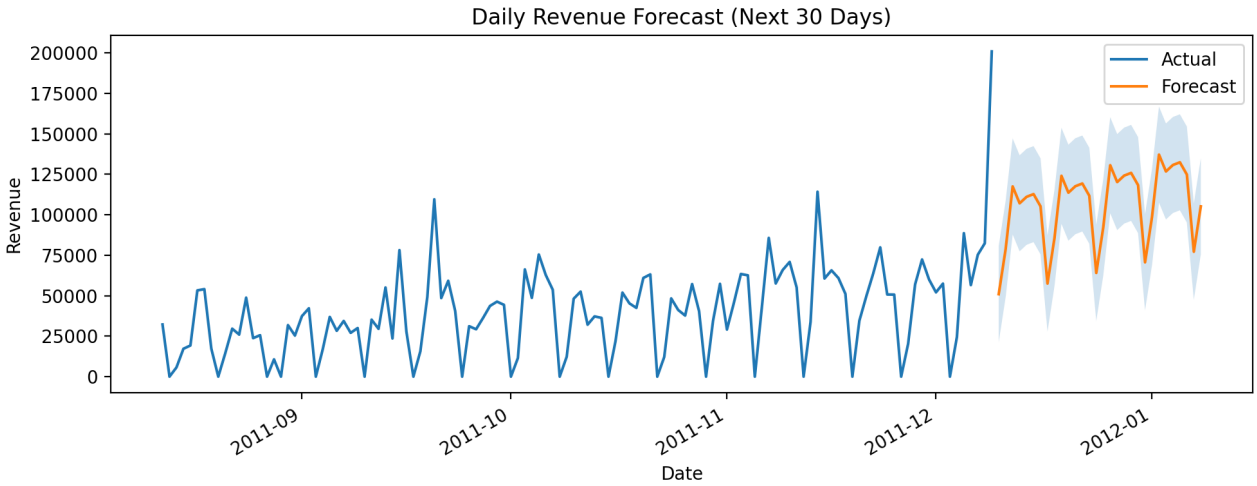
Target: Daily revenue (aggregated). Missing calendar days are filled with 0 (no sales).

Model: Exponential Smoothing (additive trend + weekly seasonality, period=7).

Backtest: last 30 days. ETS MAPE = 31.7%, RMSE = 29,918.

Baseline (seasonal naive, lag 7 days): MAPE = 24.3%, RMSE = 31,289.

Interpretation: weekly seasonality is strong; simple baselines are competitive. Use ETS for smoother forward projections, and monitor accuracy weekly.



Insights & Recommendations

• Revenue is highly concentrated in the UK: prioritize UK retention and logistics efficiency, while testing growth campaigns in the next 3-5 countries.

• Customer value is concentrated: build VIP programs for 'Champions' (personalized offers, early access) and win-back flows for 'At Risk' customers.

• Focus inventory/marketing on top-selling SKUs and bundles around seasonal peaks (notably Q4).

• Operational staffing: align fulfillment and customer support to the intra-day pattern (peak hours on weekdays).

• Forecasting: adopt a weekly revenue review cadence; compare ETS vs seasonal-naive each week and recalibrate if error rises.

Deliverables

Interactive dashboard HTML: interactive_dashboard.html

You can host the HTML on any static web server or open it locally in a browser.